He Min

China | +65-89902569
MHE009@e.ntu.edu.sg | https://qiyuanlanniao.github.io

EDUCATIONAL BACKGROUND

Nanyang Technological University — **Singapore**
Master of Science in Signal Processing and Machine Learning — 2026.01-2027.07
Chongqing University of Technology — **Chongqing, China**
Bachelor of Science in Computer Science and Technology ***(Honours)*** — 2021.09-2025.06

- **GPA:** 3.96 / 5.0 (**Top 1%**)

INTERNSHIP EXPERIENCE

Lenovo (Shanghai) Information Technology Co., Ltd. (LR Robotics Algorithm) — 2026.05-2026.08
Research Intern in Embodied AI for Robotics
Project Overview: Leveraging the Lenovo Morningstar robot as the platform, a collaborative intelligent agent system has been constructed, comprising the task-planning "mastermind" NavAgent and the dual-system navigator InternVLA-N1-DualVLN. NavAgent utilizes the LangGraph state machine and MCP to orchestrate the ROS2 toolkit, handling global task parsing, memory storage, telemetry self-checks, and anomaly diagnosis. DualVLN manages specific motion generation: System 2 (2 Hz; fine-tuned QwenVL-2.5) predicts 2D pixel sub-goals, while System 1 (30 Hz; NextDiT) outputs high-frequency physical trajectories. SR increased by 10.1%, SPL increased by 6.7%, NE decreased by 17.6%, and StR decreased by 32.5%.

PROJECT EXPERIENCE

Fine-tuning SmolVLA Robot Grasping Policies Using Multimodal Synthetic Data — 2026.07- Present
Project Overview: This project successfully executed the full embodied AI development pipeline—spanning synthetic data generation, VLA policy fine-tuning, and closed-loop evaluation—on the AMD GPU (ROCm) platform. Leveraging the Genesis physics engine, the team automatically collected and packaged a multimodal dataset of 13,500 frames in LeRobot format and performed efficient, low-memory fine-tuning on the 450-million-parameter SmolVLA model (with VRAM usage as low as 2.27 GB). Ultimately, end-to-end closed-loop control was achieved in a virtual environment, yielding a 55% success rate for grasping and lifting tasks at novel, randomly assigned target locations.

OV-LidarMap: Open-Set Instance-Level 3D Semantic Mapping with Rotating LiDAR and Panoramic Vision
Co-first Author, **IEEE RA-L (SCI Q2 TOP,** Under Review) [arXiv] Supervisor: Prof. Lihua Xie — 2026.01-2026.03
Project Overview: To address the issues of lacking spatial reasoning capabilities in Large Language Models (LLMs) and the "context explosion" problem within large-scale 3D environments, an omnidirectional perception-driven embodied AI agent framework was developed using Python, PyTorch, and ROS2. By constructing a dynamic scene graph as the agent's world model and employing a multi-resolution spatial attention mechanism, the framework achieves zero-shot long-range navigation with a Success Rate (SR) of 93.18% and a 69.98% reduction in token costs.

PUBLICATION

He, M. (2025). A Review of Data Fusion and Deep Learning Models for Multimodal Sentiment Analysis. In Proceedings of E-commerce and Artificial Intelligence (ECAI 2024). [SciTePress]

AWARDS & HONORS

National Scholarship (2024); First Prize, National College Student Mathematics Competition (2023); Third Prize, National Finals of the Team Programming Ladder Tournament (2023)

TECHNOLOGY STACK

- **Embodied Intelligence & Large Model Agent:** Proficient in **VLA (Vision-Language-Action)** end-to-end policy fine-tuning and imitation learning; skilled in **LangGraph** state machine design and encapsulation of **MCP** Protocol tool ecosystems; familiar with **CoT** reasoning chains and large model adaptive resolution / prompt compression technologies.
- **Robotic Systems, Perception and Simulation:** Proficient in **ROS 2** communication mechanisms; familiar with physics simulation engines such as **Genesis** and the **LeRobot** embodied dataset standard; experienced in **SAM2** object segmentation, **Open3D** point cloud processing and **KDTree** search; with hands-on experience in constructing 3D **dynamic scene graphs**.
- **Systems Engineering & Concurrent Programming:** Proficient in **Python** and **C++**; familiar with **Linux**, **Docker** containerization, and **Git** version control; experienced in encapsulating **low-level control logic** for **multi-threaded applications** and conducting **hardware-software integration and debugging**.